

Geometric Probability Distribution

There are four requirements for a setting to follow the geometric probability distribution. These four items include the three requirements for a Bernoulli Trial as well as one addition requirement that is unique to the Geometric Model:

- Two Possible Outcomes
 - Each trial falls into one of two categories – we call them “success” or “failure.”
 - Success does not necessarily mean a positive outcome, but instead, the outcome we are looking for.
- Fixed Probability
 - The probability of success (called p) is the same for each trial.
- Independent Trials
 - Each trial is independent of the next.
 - Reasonable Assumption for coins, cards with replacement, spinners, rolling a die, etc.
 - What about sampling without replacement, when it can't be avoided? This is where the “10% condition” comes in.
 - The 10% condition says that if you are sampling from a large enough population, you can proceed with sampling without replacement as though the trials were independent
 - Ex. Sampling from a deck of cards without replacement would violate this but sampling people from a large town would be okay.
- _____
 - This is what makes the geometric different from the binomial → what we are looking for is _____.
 - We DO NOT have a _____. We will keep going until the first success.

Examples: $X =$ _____

1. How many times do you have to flip a coin before you get tails?
2. How many times do you have to roll two dice before you get a sum of 12?
3. How many throws at the dartboard until you hit the bullseye?
4. How many people will walk through a door until a man walks in?

Before we look at the formula, let's see how we get it...

Example: You roll a 6-sided dice. Let X = number of trials until you roll a 6.

a) What is the probability that you roll a six on the first trial?
b) What is the probability that you roll another number the first trial, but a six on the second trial?
c) What is the probability that you roll another number on the first two trials, but a six on the third trial?
d) What is the probability that you roll another number of the first three trials, but a six on the fourth trial?

e) Using the four outcomes above, create a table showing this probability distribution.

X	1	2	3	4
P(X)				

Can X be zero? Why or why not? When do we stop the table? What probability pattern are you noticing?

If all of the four points above are satisfied, you can compute the number of successes in a specified number of trials:

p = probability of success x = number of successes	$P(X = x) =$	Mean: The expected number of trials until the first success. Standard Deviation: The typical deviation from the expected number of trials
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Examples

Directions: For the following situations, explain how each of the settings follows the geometric probability distribution model.

1) You toss a coin until you get a tail.	$X =$ Two Possible Outcomes: Fixed Probability: Independent Trials: First Success:	Find and interpret $P(X = 4)$
2) A company reports that 75% of emails it receives is spam.	$X =$ Two Possible Outcomes: Fixed Probability: Independent Trials: First Success:	Find and interpret $P(X = 2)$
3) Type A blood is found in 43% of the population.	$X =$ Two Possible Outcomes: Fixed Probability: Independent Trials: First Success:	Find and interpret $P(X = 5)$

Geometrics on the Calculator

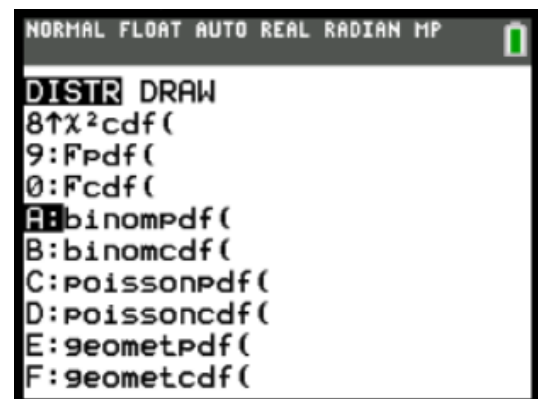
Just like with binomials, there is a difference between pdf and cdf.

_____ = probability distribution function and is used with the distribution when you are trying to calculate is $P(X = x)$

_____ can be computed using `geometpdf(p, x)`

_____ = cumulative distribution function and is used with the distribution when you are trying to calculate the probability of an interval of values

_____ can be computed using `geometcdf(p, x)`



4) Type A blood is found in 43% of the population. What is the probability that you will not find a person with Type A blood until the 3rd student walks in?

5) Type A blood is found in 43% of the population. What is the probability that you find the first person with Type A blood within 5 students?

Just like the binomial, the geometric random variable is discrete and inequalities need to be handled with the complement, just like the binomial.

6) Type A blood is found in 43% of the population. What is the probability that it will take more than 3 people before you find a person with Type A blood?